Dataverse and Large Data Support

Meeting the Challenge at Harvard Dataverse Repository

Sonia Barbosa, Leonid Andreev & Ceilyn Boyd

Meet the Team



Ceilyn Boyd
Dataverse Development Project Manager



Stefano M. Iacus
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Systems Engineer, The Dataverse Project, Data Science



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Data Technology Support Associate, Data Acquisition and Archiving, Data Science



Linda Christlieb
Finance Associate



Soner Taskaya
https://www.iq.harvard.edu/client-support-services - And the Client Support Services Team!



Sonia Barbosa
Associate Director of Dataverse Support, Data Curation, and the Murray Archive

Increased demand for large data support

- Research has become more data and compute intensive
- Increased number of requests for large data support at Harvard Dataverse
- Increased interest in computing on our data collections

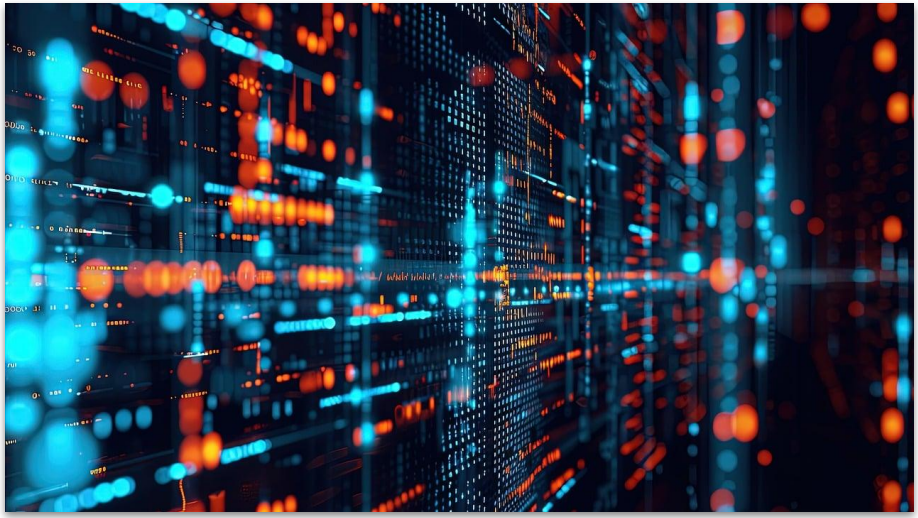


Image by Brian Penny from Pixabay

Harvard Dataverse lage data use cases

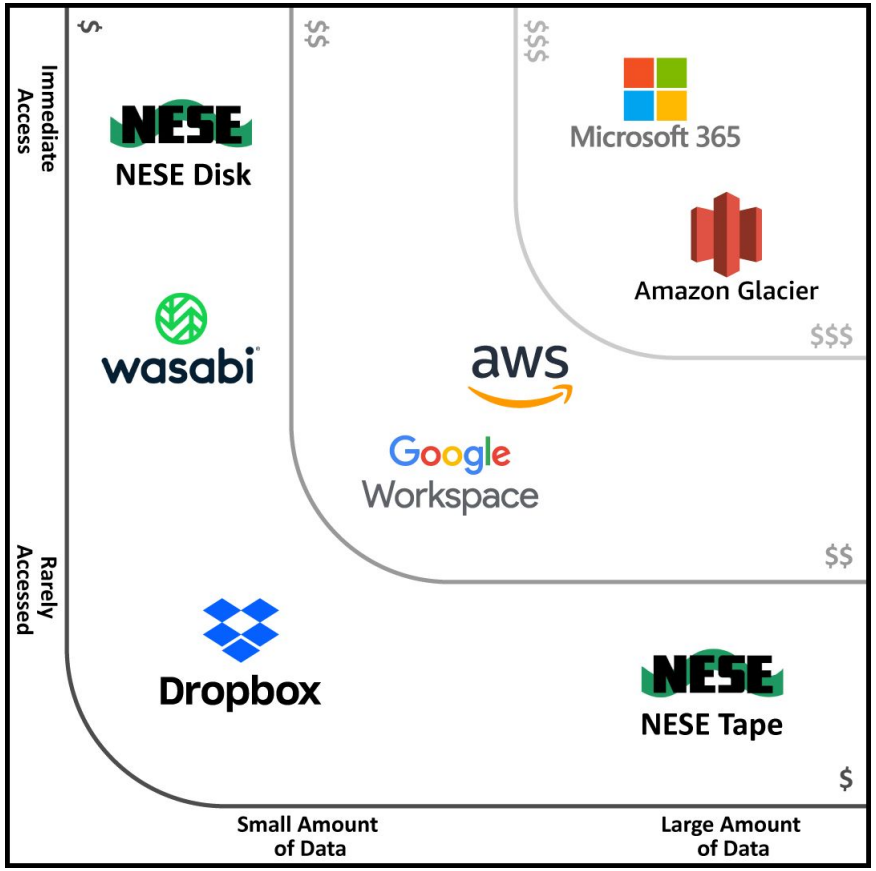
About

- 120+ tickets about large data support (5 years)
- Hospitals, Harvard & non-Harvard researchers & groups, US government dept
- 1 TB - 650 TB+ storage

Requirements

- Immediate & individual file access
- Support for millions of files
- Collection growth
- Versioning
- Restricted content
- Indefinite retention
- Budget-friendly options

Price

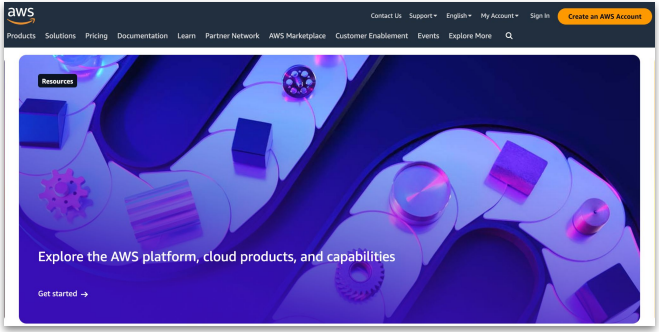


Data Access

Data Quantity

Large data support infrastructure

Amazon Web Services (AWS)/HUIT



MGHPCC

MASSACHUSETTS GREEN HIGH PERFORMANCE COMPUTING CENTER

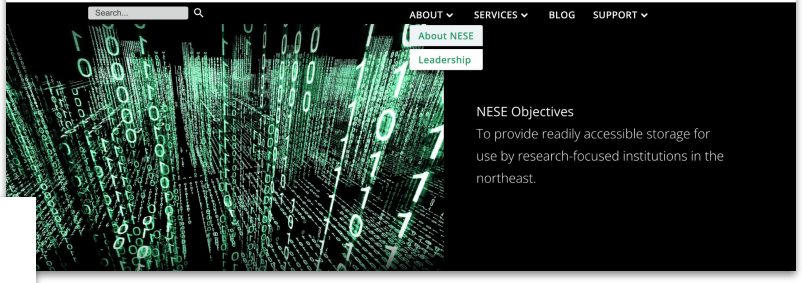
NERC

New England Research Cloud

NESE

THE NORTHEAST STORAGE EXCHANGE

Addressing the storage needs of researchers in the northeast



Northeast Storage Exchange (NESE)

Large data support infrastructure

AWS S3	NESE Disk	NESE Tape
Flexible storage w/dynamic file access	Large data storage & immediate access	Large data storage, intermediate access
• Allows dynamic access • Widely used • Admin/monitoring interface • File transfer/access costs • Expensive	• File size limitations (2T max) • File count limitations • Dynamic file access • Admin monitoring & overhead • Globus and S3/HTTP options	• File size limitations (100MB max) • File count limitations • File interactivity limitations • Lower file transfer/access costs (can be $0) • Less expensive • Admin monitoring & overhead

Harvard Dataverse large data upload & access

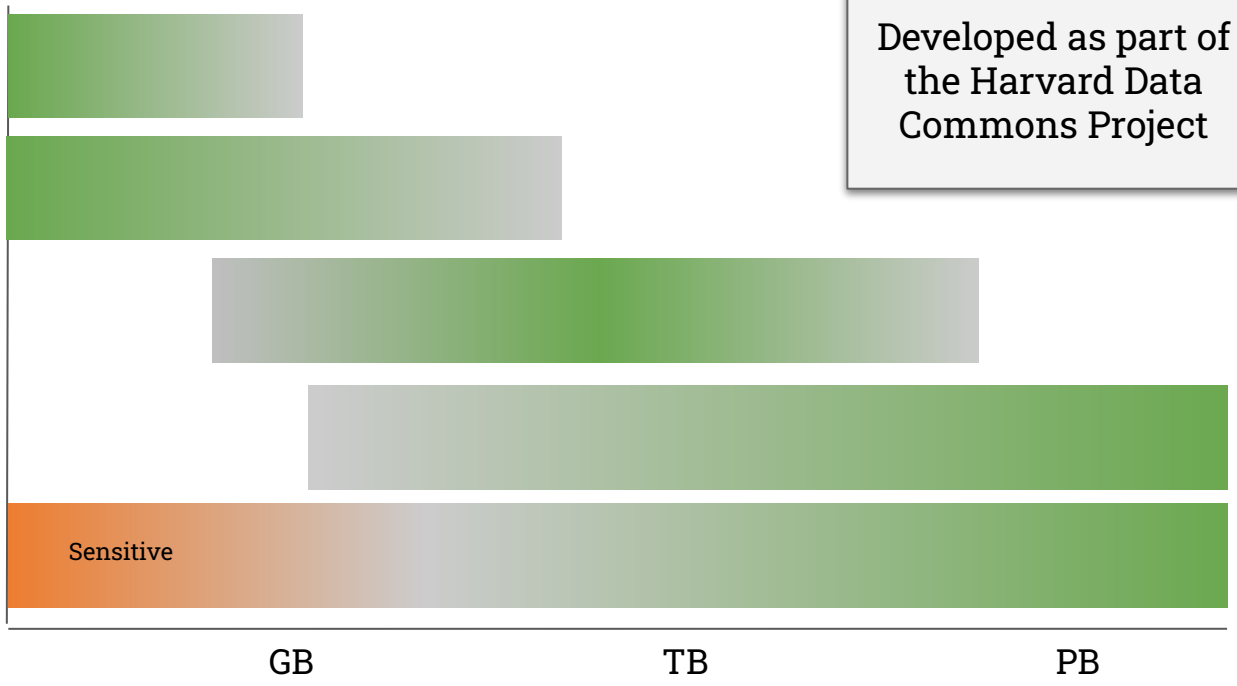
Upload through Dataverse

Direct upload/download to S3

Globus Transfer to S3

Globus Transfer to File/Tape

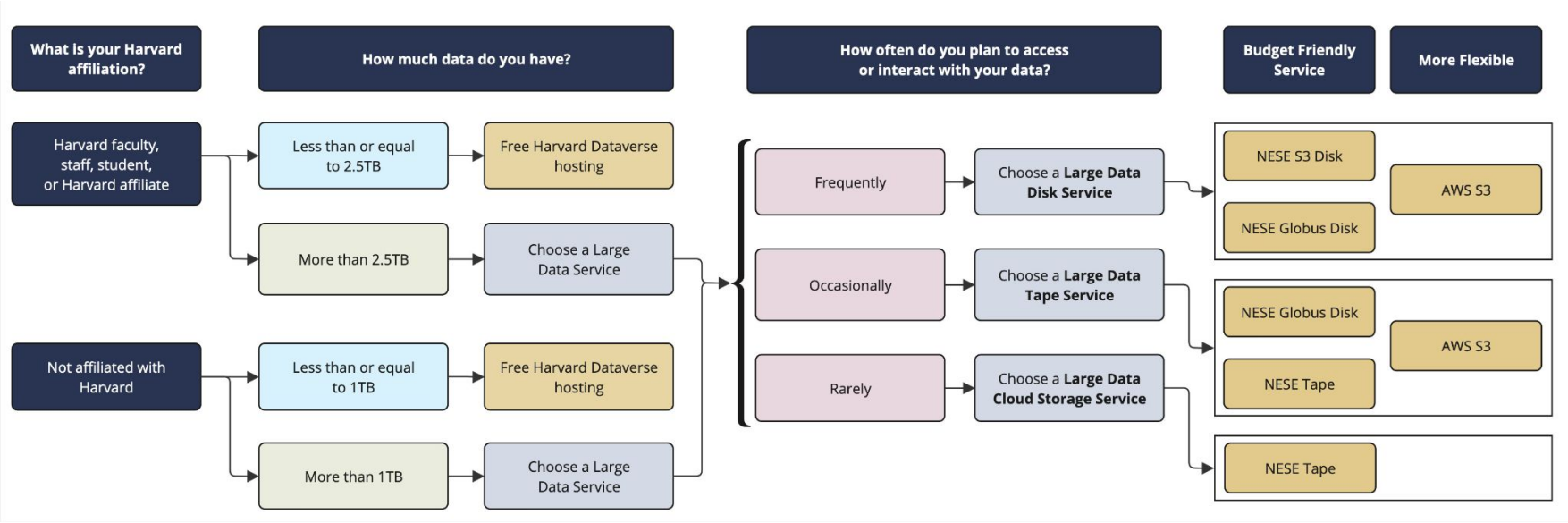
Reference Data in Remote Stores (HTTP -> Globus)



Developed as part of the Harvard Data Commons Project

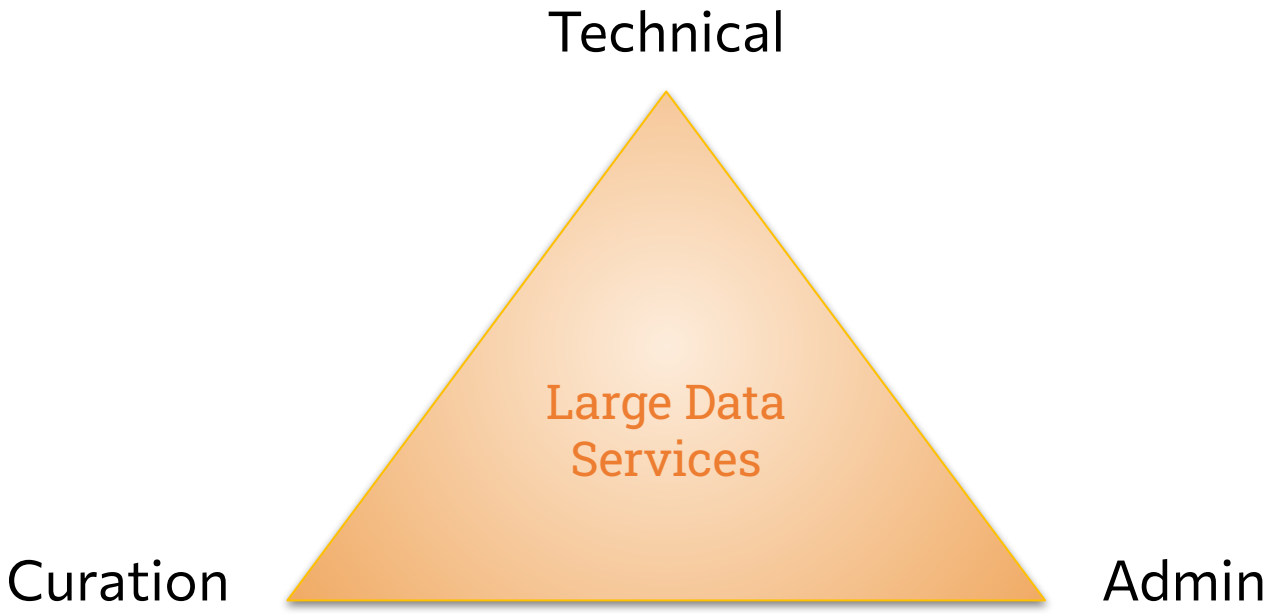
globus

When is data "large" for Harvard Dataverse?



Sonia Barbosa, Leonid Andreev & Ceilyn Boyd

Building and delivering sustainable large data services: challenges & considerations

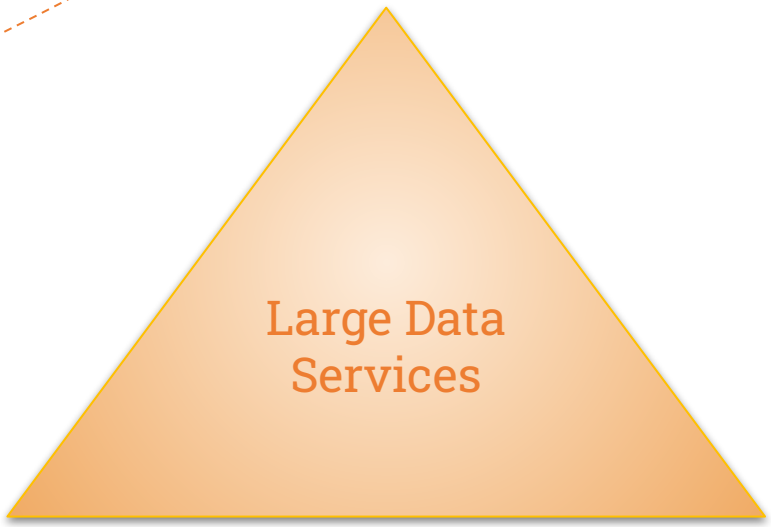


Storage & access
Data arrangement
Backups

Technical

Data representation & metadata
Data access
Special requests

Curation



Sustainability
Service coordination
Invoicing
Monitoring

Admin

Challenges and considerations: technical

Storage & access

- Data storage costs
- Data transfer costs
- File access costs
- User ↔ data and application ↔ data interaction
- Upload, downloads & computing access

Data arrangement

- File size limitations
- File count limitations

Backups

Challenges and considerations: curation

Data representation & metadata

- Custom metadata
- Metadata extraction (from existing docs)
- File arrangement

Data access

- Versioning for growing collections
- Restricted content
- Retention periods

Special requests

- Visualization, data exploration & analysis support

Challenges and considerations: admin

Sustainability

- Addressing diverse use cases w/out artisanal solutions

Cross-org collaborative service management

- Cross-org workflow coordination (e.g., support tickets, user comms, technical support)
- Revenue sharing agreements
- Account management (e.g., collecting funds from Harvard & non-Harvard groups)

Resource usage monitoring and assessment

Large data services

- Pre-Service Support
- Curation Services
- Large Data Storage & Access Services



Photo by Visual Tag Mx

Large data services workflow: participants

Large Data Services Intake Form	Consultation	Invoicing & Provisioning	Upload, Publish & Share Data
Client support	Client support	Finance	Client support
Curation	Curation	NESE	Curation
	Development support	Curation	
		Development support	

Pre-Service Services

Service Offering	Cost
Consultation (1 hour) • Consultation to describe our service offerings and recommend service offerings	• Harvard: Free • Non-Harvard: Free
Extended Consultation (2+ hours) • Extended consultation to discuss your use case and recommend service offerings	• $100/hr
Letter of Support • Preparation of a letter of support for a researcher or group planning to use Harvard Dataverse Large Data Support Services	• Harvard: Free • Non-Harvard: Free
Quote Preparation for Large Data Custom Services • Preparation of a (time-limited) quote for Large Data Custom Services only, following one or more consultations	• Harvard: Free • Non-Harvard: $200

Curation Services

Service Offering	Cost
Large Data Curation • Data curation and management support from the Dataverse Curation Team, such as metadata creation, licenses, and file layout recommendations	• See: Consultation & Extended Consultation
Large Data Training & Blog, and Conference Presentations • Webinar and blog in Winter 2025	• No cost
Dataverse Community Large Data Working Group • Community-wide discussion for installations supporting large data (Zulip Channel, Website pending)	• No cost

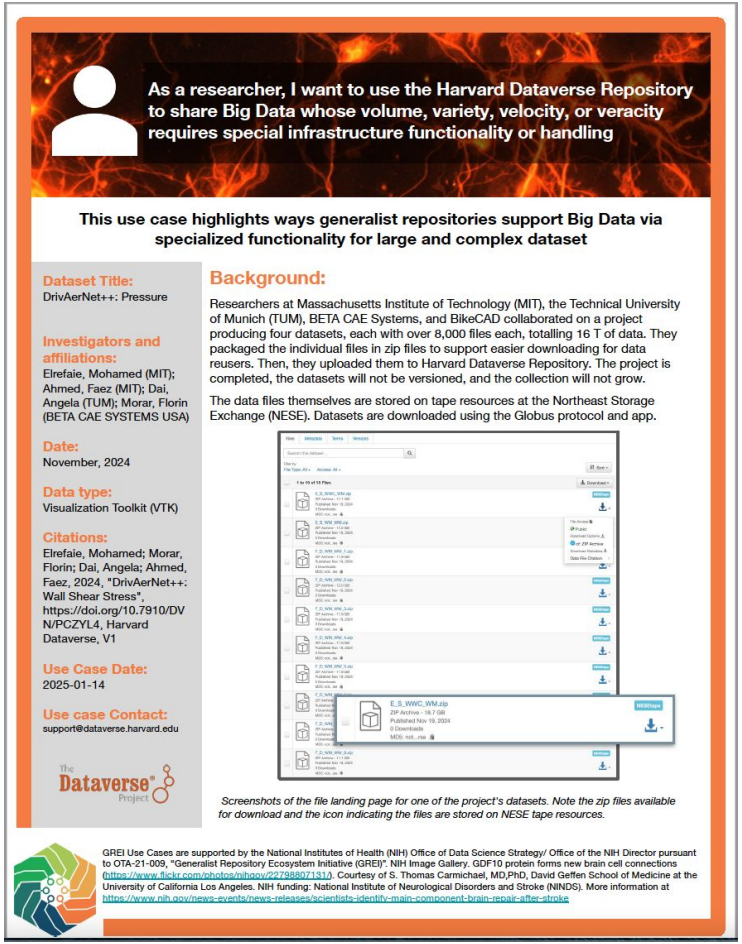
Large Data Services

Service Offering	Cost
Large Data Basic Service: Globus NESE Tape	• **Harvard:** Storage + 15% • **Non-Harvard:** Storage + 20% • 6 year min commitment
Large Data Basic Service: Globus NESE Disk	
Large Data On-Demand Service: S3 NESE Disk (forthcoming)	
Large Data On-Demand Service: AWS S3 • Separate egress costs • AWS backups	
Large Data Cold Storage: NESE Tape	
Large Data Backups • Data backups to NESE Tape	

NIH-GRE Big Data Use Case

GREI Repositories publish white paper on defining Big Data in generalist repositories and use cases showcasing approaches to Big Data support.

As the scale and complexity of data grows in the NIH data landscape, it is important that generalist repositories meet this need by offering both trademark flexibility and tailored support for sharing Big Data FAIR-ly in a trusted repository. The Generalist Repository Ecosystem Initiative (GREI) has recently published an array of documents outlining generalist repository approaches to supporting Big Data including a white paper (10.5281/zenodo.14782997) defining Big Data in the context of generalist repositories and use cases showcasing how each GREI repository supports large and complex datasets through repository functionality and services.



Sonia Barbosa, Leonid Andreev & Ceilyn Boyd

Large data services pilot and progress

Goal

- Develop & document workflows
- Establish admin infrastructure
- Create service delivery estimates
- Revise service models

Two (2) Completed Collections

- 8TB (OMAMA- SEAS)
- 16TB (DeCode Lab, MIT)

Upcoming

- 12TB - (Climate research)
- 10TB - (Marine Science)
- 650TB (SEAS)
- 120TB (Laboratory)

Large data services workflow: steps



Prospective user visits the Large Data Support Website or submits RT ticket

User completes the Large Data Services Intake Form

Prospective user has consultation/s with curation staff

Curation staff recommends best Large Data Service/s for data needs

Quote is prepared

User (ideally) accepts quote and is invoiced

Appropriate hardware is provisioned

Curation staff creates and configures HDV collection

User uploads, publishes & shares their large dataset/s

Large Data Support Website & Intake Form

https://support.dataverse.harvard.edu/large-data-support

https://harvard.az1.qualtrics.com/jfe/form/SV_78xCLVe9s8MAkQK



Large Data Support

Request Services

Request Large Data Services by completing the **Intake Form**.

About the Service

Harvard Dataverse Repository provides **affordable, large research data stewardship and archiving services** to Harvard affiliates and non-Harvard researchers and groups.

These fee-based services complement our existing Curation and Data Management Services, which include free support for research data collections up to **1.0TB for non-Harvard researchers and groups** and **2.5TB for Harvard affiliates**.

The financial contribution added to raw data storage costs helps us sustainably provide our Large Data Services users with continuing technical and administrative support and data handling, storage, and preservation costs for the duration of their service agreements.

Benefits

- Our Large Data Services benefit depositors who must publish, share, and provide long-term access to research datasets and supplementary materials with **large files or collection sizes (over 2.5TB), large numbers of files (e.g., tens of thousands or millions), or both**.
- Researchers seeking low-cost, long-term storage for smaller datasets are welcome to use our fee-based services instead of Harvard University-supported Google Drive, Dropbox, or One Drive.

Eligibility

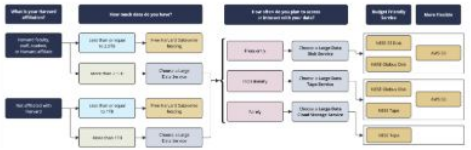
- All Harvard and non-Harvard researchers and groups.
- Fees vary depending on institutional affiliation; Harvard affiliates receive reduced-cost services.

Large Data Storage and Globus Support at Harvard Dataverse

- Large Data Services Documentation
- Readme
- Data Upload Instructions
- Data Download Instructions

Harvard Repository - Large Data Intake Request Form

Harvard Dataverse Repository provides affordable, large research data stewardship and archiving services to Harvard affiliates and non-Harvard researchers and groups. Our Large Data Services benefit depositors who must publish, share, and provide long-term access to research datasets and supplementary materials with large files or collection sizes (over 2.5TB), large numbers of files (e.g., tens of thousands or millions), or both. This service uses the Northeast Storage Exchange (NESE) and Amazon Web Services (AWS) data storage and access resources. We also partner with Harvard's FAS Research Computing and New England Research Cloud (NERC) for service delivery.



Complete the following contact information

First and Last Name:

Email address (organization/institution email):

Name of Research Project:

University affiliation:

Briefly describe your project:

Briefly describe your project, including the funding agency:

Estimated total project size in TB:

Total number of files in project:

Estimated # of years of expected storage/sharing support (minimum of 6 years):

What made you want to share your data with the Harvard Dataverse Repository?

Describe any conditions that prevent data repackaging:

List any other stakeholders associated with this project:

Do you require managed access to your data?

- Yes
- No

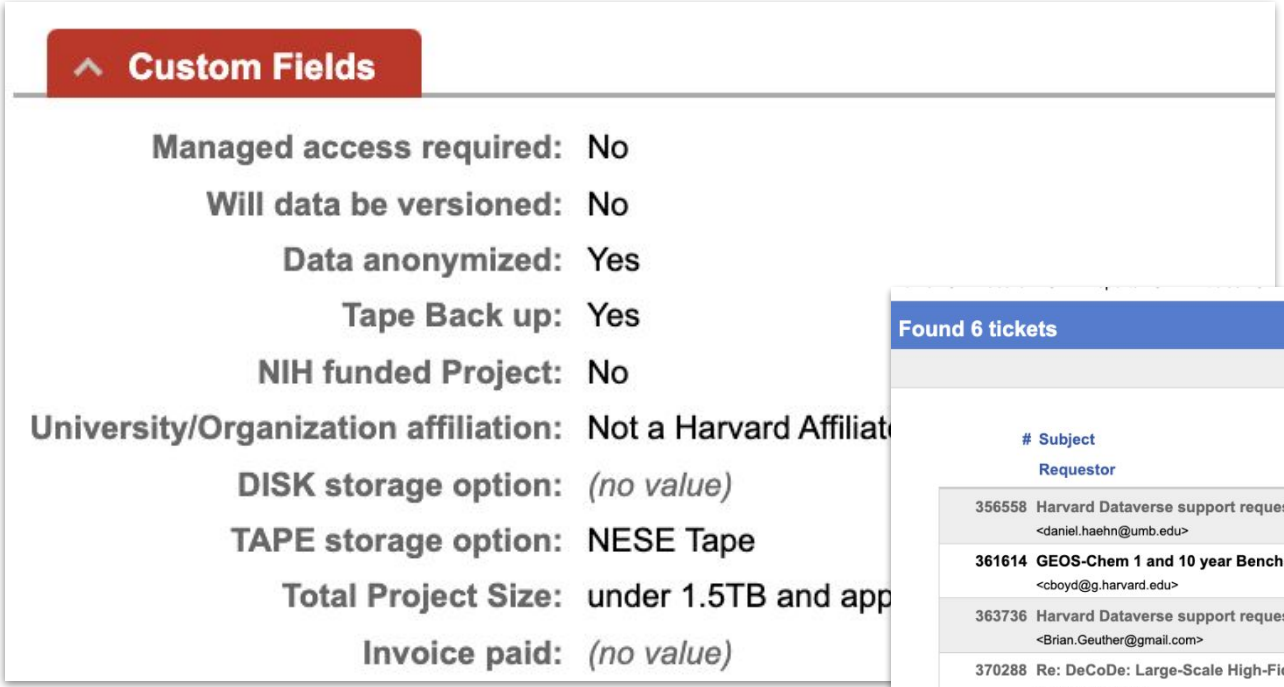
Will this data be versioned during the sharing/storage timeline indicated?

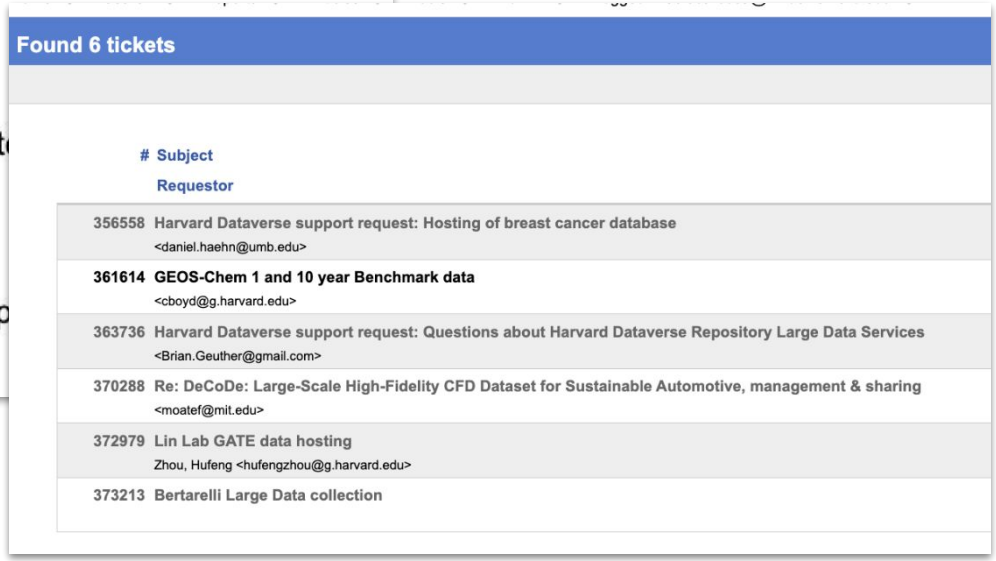
- Yes
- No

Are the data anonymized/deidentified?

- Yes
- No

Sonia Barbosa, Leonid Andreev & Ceilyn Boyd





RT Support PM for Large Data Requests

File Packaging for Large Data Support

File Access, Size, and File Count Limitations

	NESE Tape	Globus NESE Disk	S3 NESE Disk	AWS S3
Access	Globus	Globus	S3/HTTP	S3/HTTP
File Size				
Minimum	100 MB	-	-	-
Maximum	-	2 TB	1 TB	5 TB
Preferred	1 GB - 1 TB	-	-	-
File Count	10 per GB	256 per GB	256 per GB	-

Globus-specific User Documentation

If you are considering Large Data Services at HDV, we strongly recommend looking at the detailed instructions that we provide for data depositors and end users.

https://github.com/IQSS/dataverse.harvard.edu/blob/master/doc/globus/README.md

Note specifically the Quick Start download instructions and the extended separate guides for Download and Upload.

Globus-specific Workflow

- The screenshots in the next few slides are from the end user guides referenced in the last slide
- They are to illustrate the unique nature of the workflow of uploading and downloading data when using Globus-accessible storage.
- It is important to keep in mind that it is different and more complex than the workflow to which Dataverse users are normally accustomed. For example, the extra UI component.

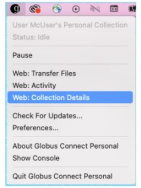
Harvard Repository - Depositing Data via Globus Integration

c. Globus Connect Personal

Globus Connect Personal is a free client offered by Globus. Linux, Mac and Windows versions are available. Once installed on your computer, Globus Connect Personal will make your local files visible to the Globus network as a collection that can be used as the data transfer source. It will be limited initially to your home directory, but any other directory or volume local to your system can be added. "Local" in this context means files directly stored on your laptop, but also any external hard drives or any remote volumes (NFS, etc.) mounted over the network.

In order to complete the installation of Connect Personal, you will be asked to log in using your Globu

You can verify that your new collection is properly set up by clicking the "Web: Collection Details" in t



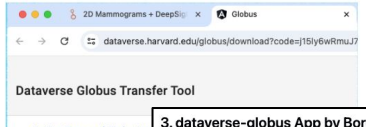
This should take you to the Globus main Web App page, confirming that your personal collection is n



Note that the above is only suggested as a way to verify that Connect Personal is working properly. You will not be using the Web App at globus.org to upload the data, all the transfers will have to be initiated from the Dataverse pages.

If necessary, add any extra folders containing the data you are depositing to the collection in the *Access* panel under *Preferences*. Since you are transferring data *from* your collection, it is safe to leave the "Write Access" box unchecked. The "Shareable" box does not need to be checked either, unless you actually need to make your personal collection accessible to other Globus users.

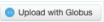
The dataverse-globus app will inform you if Globus Connect isn't installed or properly configured on your local system, giving you an option to select a different Globus collection:



You do not have Globus Connec
Please either start your Connect
select 'Active Recently Viewed E
or search for active endpoints.

3. dataverse-globus App by Borealis/GDCC

The upload process relies on an extra user interface component, "dataverse-globus", as an add-on to the Dataverse pages you may be familiar with already. Clicking the "Upload with Globus" on the *Upload Files* page in the Dataverse UI will redirect you to dataverse-globus in a new tab:

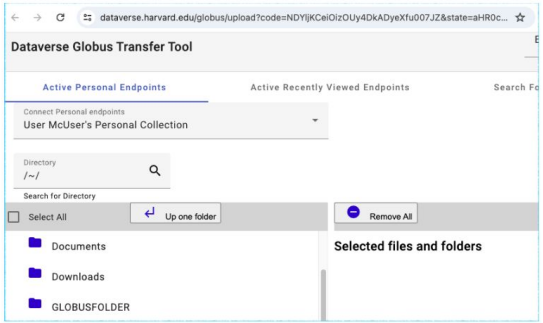


(If you are not seeing the "Upload with Globus" button above on the uploads page, that likely means that your Dataverse Collection has not yet been configured to use NESE storage - please contact support to have that straightened out)

Unless you are logged in already, you will be redirected to the Globus login page in the "dataverse-globus" tab (make sure you are using the same account that you used to configure your instance of Globus Connect Personal):



Otherwise the app will show two panels, source on the left, and the destination on the right:



dataverse-globus was originally developed by Victoria Lubitch at Borealis (formerly Scholars Portal), the Canadian Dataverse Repository project. **Note that this add-on does NOT need to be installed locally by the user; it is an additional server component on the Dataverse side.**

4. Initiating Globus Upload

Navigate to the file you want to upload in the left panel and check the box next to it. It should now appear on the right, in the *destination* panel:



then click *Submit Transfer* at the bottom of the page:



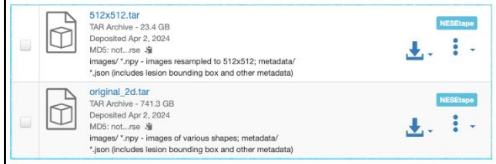
Harvard Repository - Downloading Data via Globus Integration

Downloading data from NESE via Globus: Quick Start

This is a simplified instruction on how to download files from NESE to your personal computer. A more detailed guide is available with information on extra use cases, troubleshooting and more.

1. NESE Large Data Storage at Dataverse

Files stored on remote Large Data volumes at NESE are identified on Dataset pages by an extra blue label with t
the following example the two files shown are stored on tape at NESE:



Data at NESE storage can be accessed using Globus transfer protocol (no HTTP access is available). This requi
installed on your computer:

2. Globus Connect Personal

Globus Connect Personal is a free client offered by Globus. Linux, Mac and Windows versions are available.

In order to complete the installation you will be asked to log in as a Globus user. You will likely be able to log in u
institution's federated login. If this is not an option, if your institution does not appear in the list of participating
create a Globus Id at globusid.org.

Once Globus Connect Personal is installed, you will need to make sure it is started before you can initiate any G
Dataverse.

3. Initiate the Download

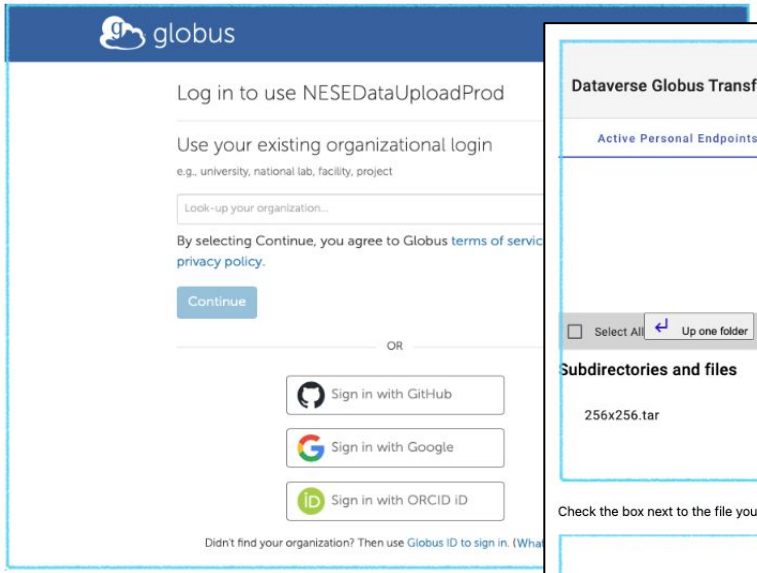
Downloads can only be initiated from dataset pages at Harvard Dataverse. (i.e. these files cannot be accessed
that bypasses Dataverse).

To download a single file, open the *File Access* menu and click on the blue *Globus Transfer* icon:



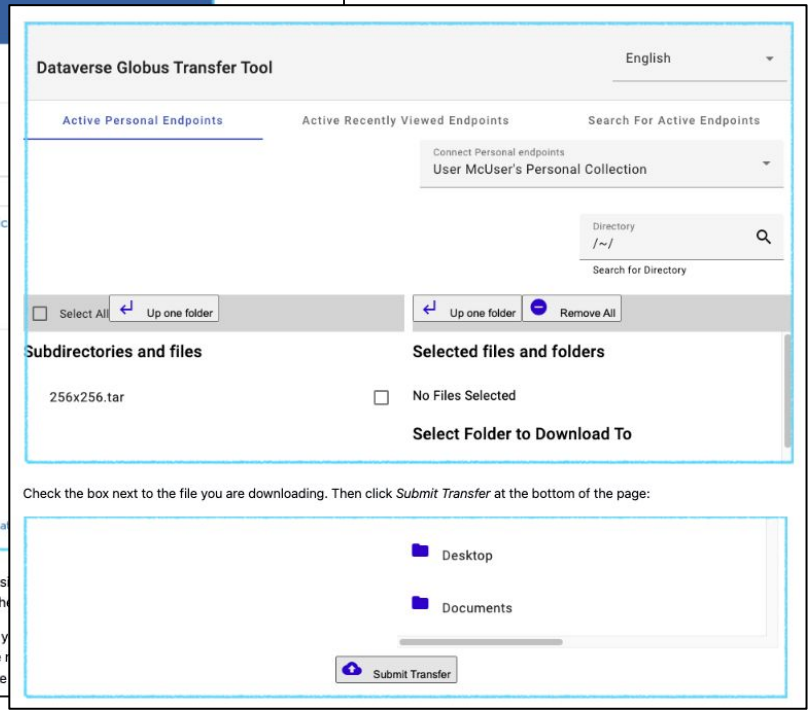
Dataverse will then redirect you to the dedicated Globus user interface ("dataverse-globus") in a new tab.

Unless you are already logged in, you will be further redirected to the Globus login page. Make sure you are using the same Globus account that you used to configure your instance of Globus Connect Personal:



You will see an error message if your Globus Connect Personal isn't running. Then s
and try again. The application should be in your Applications folder on the MacOS or th

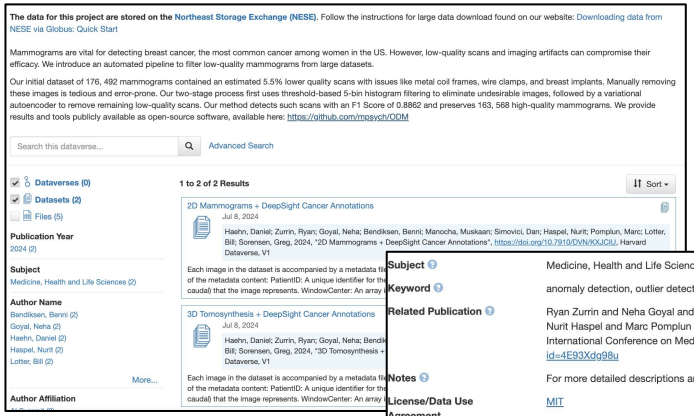
The Globus webapp should now present you the following view with 2 panels. The file y
left. Your local home directory should appear as the "destination" on the right: (please r
initialize! if you are not seeing the files you had selected on the left, please give it some



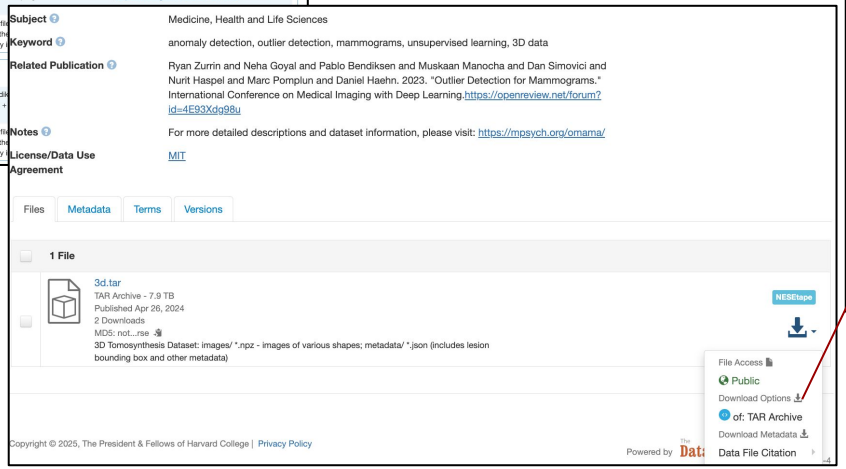
Check the box next to the file you are downloading. Then click *Submit Transfer* at the bottom of the page:

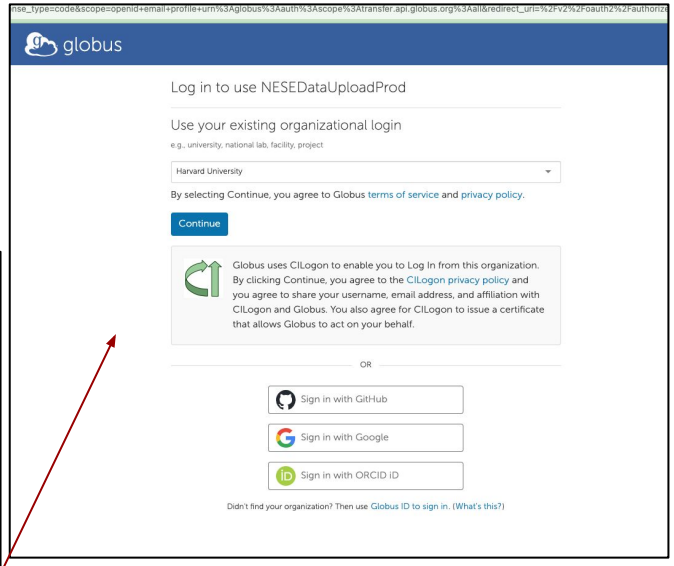
Harvard Repository Use Cases

Guestbook utilization

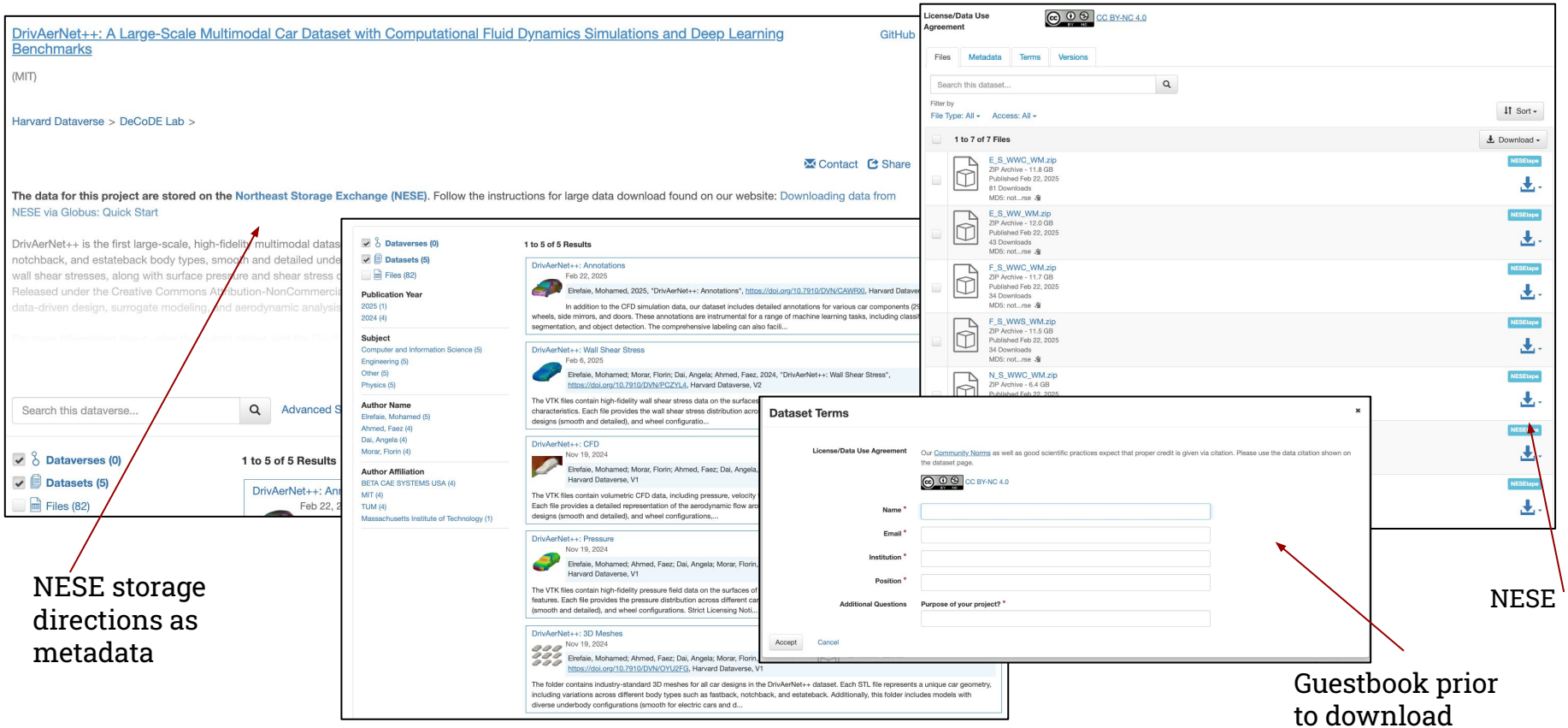








Harvard Repository Use Cases



NESE storage directions as metadata

NESE

Guestbook prior to download

Benefits and Limitations of Globus

- Once again, it is important to understand both the advantages and limitations, for you as a depositor and your end users
- The single most important advantage is that Globus makes it possible to transfer Terabyte-sized amounts of data that would be impractical to upload or download via HTTP
- One drawback is the extra complexity and extra components
- Another, possibly biggest disadvantage is that all transfers must be initiated in the UI. Even for publicly available data, there is no option of issuing an anonymized access URL.

Live Demo

References

GREI repositories and Big Data

Large Data Support in Harvard Repository

Harvard Repository Globus Guide to deposits and downloads
Harvard Repository Large Data Request Intake Form



Q & A